Do Now · retrieval

Scope, modularity and parameters

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

If procedural Deposit(balance,5) returns a value that the caller ignores, does balance change?

Q2

What is a local variable?

Q3

How does a value-returning function pass a result back?

Q4

Why might using ref unexpectedly make a method harder to debug?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Copying an integer — Assignment b=a copies the integer 5, not a reference to a storage location.

Core idea 2

Passing by reference — Increase(ref a) gives parameter value an alias of a, so the caller a changes to 8.

Core idea 3

Use modular methods — A short reusable method has a clear input effect, can be tested, and reduces repeated code.

Key visual · complete concept

AFTER RETURN

VARIABLE	VALUE / REFERENCE
a	8
b	5
ref value	out of scope

a and b now differ.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

scope — Where a name can be accessed.

local — Variable with block or method-level scope.

global/class-level — Variable accessible in a wider scope, subject to access rules.

by value — Pass a copy of the value.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

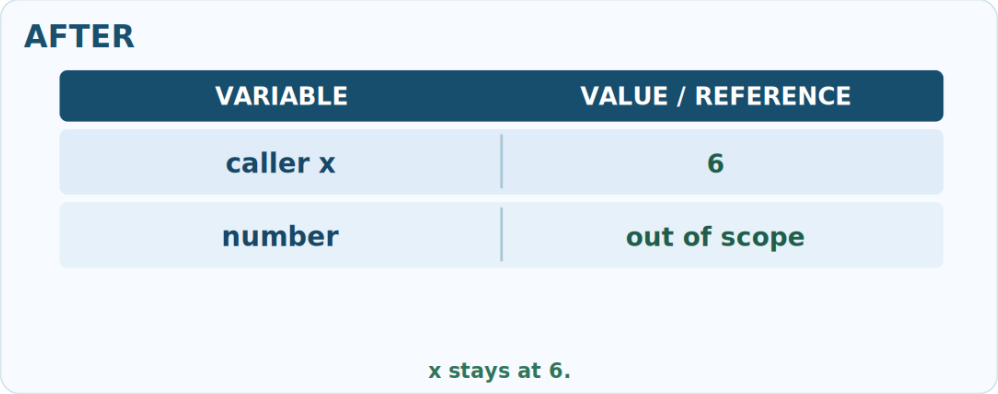
Problem

Trace AddFour(x) by value when caller x starts at 6.

Follow the worked method

1. Predict first — The function parameter number receives a copy of 6.
2. Model and check — Inside the function number becomes 10; caller x remains 6.
3. Model and check — The parameter leaves scope at the end of the call.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The ref parameter number refers to y, not a fresh copy.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Method AddTwo(int n) adds 2 to n. Caller x=7 is passed by value. What is x after the call?

Question 2

Write an OCR-style subroutine Swap with two by-reference integer parameters and explain why by-reference is required.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare `int result=Double(x);` with `Double(ref x);` as ways of updating a number. Give advantages of one method for a pure calculation.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Passing an integer by value in C# changes the caller's integer when the parameter changes.

Use key vocabulary from memory

1. Explain scope in your own words.

2. Explain local in your own words.

3. Explain global/class-level in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.